

Chapter 19 was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

CHAPTER 19

CONCRETE

SECTION BC 1901 GENERAL

1901.1 Scope. The provisions of this chapter shall govern the materials, quality control, design and construction of concrete used in structures.

1901.1.1 Referenced standards. Where this code makes reference to the nationally recognized standard ACI 318, such standard shall be as modified for New York City in accordance with Section 1908 of this code. Where sections within ACI 318 are referenced in other chapters and appendices of ACI 318, the provisions of this chapter shall apply.

1901.2 Plain and reinforced concrete. Structural concrete shall be designed and constructed in accordance with the requirements of this chapter and ACI 318 as amended in Section 1908 of this code. Except for the provisions of Sections 1904 and 1910, the design and construction of slabs on grade shall not be governed by this chapter unless they transmit vertical loads or lateral forces from other parts of the structure to the soil. Precast concrete diaphragms in buildings assigned to Seismic Design Category C or D shall be designed in accordance with the requirements of ASCE 7, Section 14.2.4.

1901.3 Anchoring to concrete. Anchoring to concrete shall be in accordance with ACI 318 as amended in Section 1908, and applies to cast-in (headed bolts, headed studs and hooked J- or L-bolts), post-installed expansion (torque-controlled and displacement-controlled), undercut and adhesive anchors.

1901.4 Composite structural steel and concrete structures. Systems of structural steel acting compositely with reinforced concrete shall be designed in accordance with Section 2206.

1901.5 Construction documents. The construction documents for structural concrete construction shall include:

1. The specified compressive strength of concrete at the stated ages or stages of construction for which each concrete element is designed.
2. The specified strength or grade of reinforcement.
3. The size and location of structural elements, reinforcement and anchors.
4. Provision for dimensional changes resulting from creep, shrinkage and temperature.
5. The magnitude and location of prestressing forces.
6. Anchorage length of reinforcement and location and length of lap splices.
7. Type and location of mechanical and welded splices of reinforcement.
8. Details and location of contraction or isolation joints specified for plain concrete.
9. Minimum concrete compressive strength at time of posttensioning.
10. Stressing sequence for posttensioning tendons.
11. For structures assigned to Seismic Design Category D, a statement if slab on grade is designed as a structural diaphragm.
12. Exposure classes per ACI 318, Section 19.3.1, and Section 1904.2 of this code.
13. Areas of mass-concrete where a thermal control plan is required.

1901.6 Special inspections and tests. Special inspections and tests of concrete elements of buildings and structures, and concreting operations, shall be as required by Chapter 17.

CONCRETE

SECTION BC 1902 DEFINITIONS

1902.1 Definition.‡ The following term is defined in Chapter 2:

MASS CONCRETE.

SECTION BC 1903 SPECIFICATIONS FOR TESTS AND MATERIALS

1903.1 General. Materials used to produce concrete, concrete itself and testing thereof shall comply with the applicable standards listed in ACI 318 and this section.

1903.2 Special inspections. Where required, special inspections and tests shall be in accordance with Chapter 17.

1903.3 Glass fiber-reinforced concrete. Glass fiber-reinforced concrete (GFRC) and the materials used in such concrete shall be in accordance with the PCI MNL 128 standard.

1903.4 Flat wall insulating concrete form (ICF) systems. Insulating concrete form material used for forming flat concrete walls shall conform to ASTM E 2634.

1903.5 Water. Water used in mixing concrete shall be clean and free from injurious amounts of oils, acids, alkalis, salts, organic materials or other substances that are deleterious to concrete or steel reinforcement and shall comply with ACI 318, Section 26.4.1.3.

1903.6 Steel reinforcement. Reinforcement and welding of reinforcement to be placed in concrete construction shall conform to the requirements of ACI 318, Chapter 20, and this section.

1903.6.1 Reinforcement type. Steel reinforcement used in concrete construction shall comply with ACI 318, Chapter 20.

1903.6.2 Welding. Welding of reinforcing bars shall conform to AWS D1.4 and ACI 318, Section 26.6.4. Type and location of welded splices and other required welding of reinforcing bars shall be indicated on the construction documents or in the project specifications. The ASTM reinforcing bar specifications, except for ASTM A 706, shall be supplemented to require a report of material properties necessary to conform to the requirements in AWS D1.4. A written welding procedure shall be provided to the registered design professional of record for approval prior to any welding. All welding shall be subject to special inspection by a special inspection agency.

1903.7 Admixtures. Admixtures to be used in concrete shall be subject to prior approval by the registered design professional of record and shall comply with ACI 318, Section 26.4.1.4. Admixtures shall be included in the concrete design mix submittals.

1903.8 Storage of materials. The storage of materials for use in concrete shall comply with ACI 318, Section 26.5.1, and the provisions of Sections 1903.8.1 and 1903.8.2 of this chapter.

1903.8.1 Manner of storage. Cementitious materials and aggregates shall be stored in such a manner as to prevent deterioration or intrusion of foreign matter.

1903.8.2 Unacceptable material. Any material that has deteriorated or has been contaminated shall not be used for concrete.

SECTION BC 1904 DURABILITY REQUIREMENTS

1904.1 General. Concrete shall be designed and proportioned to comply with the durability, strength, exposure, and other requirements as specified in ACI 318 and Sections 1904.2 through 1904.5 of this chapter.

1904.2 Exposure categories and classes. Concrete shall be assigned to exposure classes in accordance with ACI 318, Section 19.3.1, based on:

1. Exposure to freezing and thawing in a moist condition or exposure to deicing chemicals;
2. Exposure to sulfates in water or soil;
3. Exposure to water where the concrete is intended to have low permeability; and
4. Exposure to chlorides from deicing chemicals, salt, saltwater, brackish water, seawater or spray from these sources, where the concrete has steel reinforcement.

1904.3 Concrete properties. Concrete mixtures shall conform to the most restrictive requirements of ACI 318, Section 19.3.2, based on the exposure classes assigned in Section 1904.2 of this chapter.

Exception: For occupancies and appurtenances thereto in Group R occupancies that are in buildings less than four stories above grade plane, normal-weight aggregate concrete is permitted to comply with the requirements of Table 1904.3 of this chapter in lieu of the requirements of ACI 318, Table 19.3.2.1.

TABLE 1904.3
MINIMUM SPECIFIED COMPRESSIVE STRENGTH (f'_c)

TYPE OR LOCATION OF CONCRETE CONSTRUCTION	MINIMUM SPECIFIED COMPRESSIVE STRENGTH (f'_c at 28 days, psi)
Basement walls, foundation walls, exterior walls and other vertical concrete surfaces, basement slabs, interior slabs on grade except for garage floor slabs, and foundations not exposed to the weather	3,000 ^a
Driveways, curbs, walks, patios, porches, carport slabs, steps and other flatwork exposed to the weather, and garage floor slabs	3,500 ^b
Structural concrete exposed to weather	4,500

For SI: 1 pound per square inch = 0.00689 MPa.

- a. Concrete in these locations that can be subjected to freezing and thawing during construction shall be of air-entrained concrete in accordance with Section 1904.2.
- b. Concrete shall be air entrained in accordance with Section 1904.4.1.

1904.4 Freezing and thawing exposures. Concrete that will be exposed to freezing and thawing, in the presence of moisture, with or without deicing chemicals being present, shall comply with ACI 318, Section 19.3.3, and Sections 1904.4.1 and 1904.4.2 of this code.

1904.4.1 Air entrainment. Concrete exposed to freezing and thawing while moist shall be air entrained in accordance with ACI 318, Sections 19.3.3.1 and 19.3.3.3.

1904.4.2 Deicing chemicals. For concrete exposed to freezing and thawing in the presence of moisture and deicing chemicals, the maximum weight of fly ash, other pozzolans, silica fume or slag that is included in the concrete shall not exceed the percentages of the total weight of cementitious materials permitted by Table 1904.4.2.

TABLE 1904.4.2
REQUIREMENTS FOR CONCRETE EXPOSED TO DEICING CHEMICALS

CEMENTITIOUS MATERIALS	MAXIMUM PERCENT OF TOTAL CEMENTITIOUS MATERIALS BY WEIGHT ^{a, b}
Fly ash or other pozzolans conforming to ASTM C 618	25 ^d
Slag conforming to ASTM C 989	50
Silica fume conforming to ASTM C 1240	10
Total of fly ash or other pozzolans, slag and silica fume	50 ^c

CONCRETE

Total of fly ash or other pozzolans and silica fume	35°
---	-----

- a. The total cementitious material also includes ASTM C 150, ASTM C 595, ASTM C 845 and ASTM C 1157 cement.
- b. The maximum percentages shall include:
 1. Fly ash or other pozzolans present in Type IP or I (PM) blended cement, ASTM C 595, or ASTM C 1157.
 2. Slag used in the manufacture of an IS or I (SM) blended cement, ASTM C 595, or ASTM C 1157.
 3. Silica fume, ASTM C 1240, present in a blended cement.
- c. Fly ash or other pozzolans and silica fume shall constitute no more than 25 and 10 percent, respectively, of the total weight of the cementitious materials. The amount of fly ash or other pozzolans conforming to ASTM C 618 shall be permitted to be increased in accordance with footnote d, provided the maximum percentage of supplemental cementitious materials does not exceed the maximum values stated in Table 1904.4.2 when combining multiple types of supplemental cementitious materials.
- d. Amount of fly ash or other pozzolans conforming to ASTM C 618 in a given concrete mixture can be increased to 35% provided that the concrete is not placed during cold weather as defined by ACI 306 or that the concrete is membrane cured and protected from freezing until the concrete has attained 3,500 psi.

1904.5 Alternative cementitious materials for sulfate exposure. Combinations of cementitious materials for use in sulfate-resistant concrete that are alternative to those listed in ACI 318, Table 19.3.2.1, shall be permitted in accordance with ACI 318, Section 26.4.2.2.

SECTION BC 1905 CONCRETE QUALITY, MIXING AND PLACING

1905.1 General. The required strength and durability of concrete shall be governed by compliance with the proportioning, testing, mixing and placing provisions of Sections 1905.1.1 through 1905.13 of this code and ACI 318.

1905.1.1 Strength. Concrete shall be proportioned to provide an average compressive strength in accordance with Section 1905.3, and shall satisfy the durability criteria of Section 1904. Concrete shall be produced to minimize the frequency of strengths below f'_c in compliance with Section 1905.6.3.3. For concrete designed and constructed in accordance with this chapter, f'_c shall not be less than 2,500 psi (17.22 MPa). No maximum specified compressive strength shall apply unless restricted by a specific provision of this code or ACI 318.

1905.1.2 Cylinder tests. Requirements for f'_c shall be based on tests of cylinders made and tested in accordance with Section 1905.6.3.

1905.1.3 Basis of f'_c . Unless otherwise specified, f'_c shall be based on 28-day tests. If other than 28 days, test age for f'_c shall be as indicated in construction documents. If at any time during the concrete operations, the concrete is not in conformance with ASTM C 94 or is otherwise compromised, it shall be rejected. If it is placed, the location of the load shall be recorded and a pair of cylinders shall be molded from the load per ASTM C 31 and included in the average f'_c .

1905.1.4 Lightweight aggregate concrete. Where design criteria in ACI 318, Section 19.2.4, provides for use of a splitting tensile strength value of concrete (f'_{ct}), laboratory tests shall be made in accordance with ASTM C 330 to establish the value of f'_{ct} corresponding to the specified value of f'_c .

1905.1.5 Field acceptance. Splitting tensile strength tests shall not be used as a basis for field acceptance of concrete.

1905.2 Selection of concrete proportions. Concrete proportions shall be determined in accordance with ACI 318 and the provisions of Sections 1905.2.1 through 1905.2.4 of this code.

1905.2.1 General. Proportions of materials for concrete shall be established to provide:

1. Workability and consistency to permit concrete to be worked readily into forms and around reinforcement under the conditions of placement to be employed, without segregation or excessive bleeding.
2. Resistance to special exposures as required by Section 1904.
3. Conformance with the strength test requirements of Section 1905.6.

1905.2.2 Different materials. Where different materials are to be used for different portions of proposed work, each combination shall be evaluated.

1905.2.3 Basis of proportions. Concrete proportions shall be established in accordance with Section 1905.3 or Section 1905.4, and shall comply with the applicable requirements of Section 1904.

1905.2.4 Mass concrete. For areas of high-strength mass concrete, including but not limited to, 8,000 psi (55 158 kPa) or greater and the least dimension being equal to or greater than 36 inches (914 mm), as indicated on the construction documents, a thermal control plan complying with ACI 301, Section 8.1 and acceptable to the registered design professional of record shall be submitted by the contractor.

1905.3 Proportioning on the basis of field experience and/or trial mixtures. Concrete proportioning determined on the basis of field experience or trial mixtures shall be done in accordance with ACI 318, Sections 26.4.3 and 26.4.4, and Sections 1905.3.1 through 1905.3.5 of this chapter.

‡‡‡ **1905.3.1 Required strength.** If the required f'_{cr} is obtained for trial batch mixes prior to the date specified, the trial mix design may be approved by the registered design professional of record for the structural design.

1905.3.2 Performance and approval. Concrete mix designs shall be performed by an approved agency or by a concrete producer approved by the department as an approved fabricator in accordance with Section 1704.2.2, and approved in accordance with the requirements of Sections 1905.3.2.1 through 1905.3.2.3.

1905.3.2.1 Certification. All mix designs submitted for review and approval in accordance with Section 1905.3.2.2 shall be certified by the director of the approved agency or the quality control manager of the approved concrete producer that performed the tests used to substantiate the mix design. In addition, all mix designs shall be signed by an owner or officer of the concrete production facility.

1905.3.2.2 Review and approval. All mixes shall be approved by the registered design professional of record and filed with the department prior to use in construction.

1905.3.2.3 Distribution. Copies of the approved mix design(s) shall be maintained at the construction site by the contractor and made available to the department upon request. A copy of the approved mix design(s) shall be provided by the contractor to the licensed concrete testing laboratory performing the acceptance testing of the concrete at the construction site.

1905.3.3. Test records and trial mixtures. Field strength test records or trial mixtures used to determine f'_{cr} shall not be more than 24 months old measured from the date of batching, provided the materials are the same as those proposed for use and the concrete producer provides a written statement confirming the materials are the same as those proposed for use.

Exceptions:

1. Coarse aggregates may be acquired from an alternative source provided the nominal maximum size and the volume of coarse aggregate used remain unchanged.
2. Fine aggregates may be acquired from an alternative source provided the volume of fine aggregate used remains unchanged.

1905.3.4 Materials. The type and source of the materials used in the concrete shall be identified in the proposed concrete mix design. In order for materials in field strength test records or trial mixtures to be considered the same as those proposed for use, the materials shall, at a minimum, comply with the following:

1. Cementitious materials. Cementitious materials shall be used in the same quantities for each individual component and shall be of the same type and shall be from the same source and mill. Upon request, the concrete producer shall provide mill certificates for the cementitious materials proposed for use and cementitious materials represented by the field strength test records or trial mixtures to the registered design professional of record and the department.
2. Aggregates. Coarse and fine aggregates shall be the same nominal maximum size and shall be used in the same volume for each individual component. The aggregate supplier shall attest that the aggregates proposed are not alkali-silica reactive. The aggregate supplier is permitted to rely on New York State Department of Transportation approval of the supplier's aggregate source.

CONCRETE

3. Admixtures. Admixtures shall meet the requirements of the same admixture type in accordance with ASTM C 494 or ASTM C 260 and there shall be no change in air content or water content from the previously specified mix.

Exception: Addition or removal of retarding or accelerating admixtures as described in Section 1905.3.5 is acceptable.

4. Additional testing. Differences or changes in materials that are proposed for use, not meeting the requirements of Items 1 through 3 above, shall require new testing demonstrated by either field strength test records or trial mixtures.

Exception: Field changes to concrete proportions specified in Section 1905.3.5.

1905.3.5 Field changes to concrete proportions. Based on test data and observations during the course of construction, the following changes described in Sections 1905.3.5.1 and 1905.3.5.2 are permitted without requiring additional trial mixtures or field strength test records, provided:

1. There are no changes to the type and source of materials described in Sections 1905.3.3 and 1905.3.4, except as permitted in Sections 1905.3.5.1 and 1905.3.5.2;
2. Documentation of these changes are maintained by the concrete producer and the concrete contractor;
3. The changes described in Section 1905.3.5 cannot be used for the submission of an initial mix design to demonstrate that a previously prepared mix design meets strength or air content requirements that differ from the required concrete properties without new field strength test records or new trial mix designs;
4. Where required in Section 1905.3.5, documentation of these changes shall be submitted for acceptance to the registered design professional of record for the structural design by the concrete producer prior to use;
5. Copies of the revised mixtures shall be maintained at the construction site and provided to the licensed concrete testing laboratory performing the acceptance testing of concrete in accordance with Item 4 of Table 1705.3 of this code; and
6. The documentation required by Items 2 and 5 above shall be made available to the commissioner upon request.

1905.3.5.1 Changes that do not require prior approval of the registered design professional of record for the structure. The following admixture modifications to concrete proportions can be made without requiring prior acceptance by the registered design professional of record:

1. Air-entraining admixtures. Air-entraining admixture dosages can be increased or decreased as needed to obtain the air content specified in the construction documents.
2. Water reducing admixtures. Water reducing admixture dosages can be increased or decreased to produce the specified workability, provided that the design water is not exceeded and the changes do not segregate the concrete mix.

1905.3.5.2 Changes that require prior approval of the registered design professional of record for the structure. The following modifications to concrete proportions can be made only with prior acceptance by the registered design professional of record. Any added materials and the final mix proportions shall be indicated in a revised mix design submitted to the registered design professional of record prior to use:

1. Retarding and accelerating admixtures. Retarding and accelerating admixtures may be added to concrete mixtures or the dosage may be modified as needed to maintain practical set times.
2. Pigment. Pigment dosage may be adjusted to maintain uniform concrete color.
3. Fibers. Synthetic, glass or natural fibers may be added to the mix within the manufacturer recommended range. Any loss in workability shall be compensated with a water reducer.

Exception: Steel fibers may not be added to a concrete mixture without the performance of a new mix design.

4. Corrosion inhibitor. Corrosion inhibiting admixtures may be added, provided the corresponding adjustment of mix water remains within the limits of the required water/cementitious material ratio. The added

materials and the revised design mix shall be submitted for review by the registered design professional of record for the structure prior to use.

5. Cementitious content and water-cementitious material ratio (w/cm). The cementitious content may be increased by a maximum of 12 percent, or the w/cm ratio may be decreased while maintaining the same cementitious content, but the w/cm and air content shall not be increased. Aggregate quantities shall be adjusted, as required, to maintain the mix proportion and yield.
6. Hydration control. Admixture may be added for the purpose of delaying hydration and maintaining a workable mix.
7. Source of aggregate. Change to the source of aggregate may be approved by the registered design professional of record without providing additional test data only when the original aggregate source is no longer producing the required aggregates or the quality of the aggregates produced no longer meets the requirements of ASTM C 33 or the New York State Department of Transportation, provided:
 - 7.1. The new aggregate source is approved by the New York State Department of Transportation;
 - 7.2. A modulus of elasticity is not specified;
 - 7.3. The volume of both the fine and coarse aggregates remains the same and the nominal maximum size of coarse aggregates is unchanged;
 - 7.4. Lightweight aggregate shall not be substituted for dense, nor dense aggregate for lightweight, where lightweight aggregates are defined as having a specific gravity less than 2.5; and
 - 7.5. The registered design professional of record is informed of the reason necessitating the change of aggregate.
8. Cementitious materials. Any change to cementitious materials shall require an additional laboratory testing trial batch. The new cementitious material shall maintain the same air and water cementitious ratio (w/cm) as the original mix and meet the strength requirements of ACI 301, Table 4.2.3.3 b. Approval of the registered design professional of record and submittal of data from either a three- or seven-day compressive strength test shall be required prior to the placement of concrete.

1905.4 Proportioning without field experience or trial mixtures. Concrete proportioning determined without field experience or trial mixtures shall be done in accordance with ACI 318, Section 26.4.4.1(b). This method of proportioning shall not be permitted for:

1. Load-carrying structural concrete where the total volume of concrete on a given project exceeds 50 cubic yards (38 m³); or
2. Structural or non-structural concrete mix proportions that will be exposed to exposure classes S1, P1, or C2.

1905.5 Average strength reduction. As data becomes available during construction, it is permissible to reduce the amount by which the average compressive strength (f'_c) is required to exceed the specified value of f'_c in accordance with ACI 318, Section 26.4.4.1(c). A new mix design shall be submitted to the registered design professional of record for approval prior to use.

1905.6 Evaluation and acceptance of concrete. The criteria for evaluation and acceptance of concrete shall be as specified in ACI 318, Sections 26.12 and 26.13, and Sections 1905.6.1 through 1905.6.5 of this code.

1905.6.1 Acceptance testing. Concrete shall be tested in accordance with the requirements in Sections 1905.6.2 through 1905.6.5. Concrete sampling and testing for acceptance shall be deemed a special inspection performed by a licensed concrete testing laboratory as an approved agency. An approved agency shall perform tests on fresh concrete at the job site, prepare specimens required for curing under field conditions, prepare specimens required for testing in the laboratory and record the temperature of the fresh concrete when preparing specimens for strength tests. All field sampling and testing, including the testing of aggregates, concrete mixes, and strength testing of specimens, shall be subject to special inspection by an approved agency. All testing laboratories shall be approved agencies and shall employ qualified special inspectors to perform all required field and laboratory tests. Test results shall be promptly distributed by the testing laboratory to the registered design professional of record, concrete producer, owner and contractor to allow for corrective action where the concrete is found to be noncompliant with the mix design or noncompliant with the requirements otherwise specified in this code. Unless more prompt action is required, test results shall be reported no more than 7 business days after the date the respective tests were performed.

CONCRETE

1905.6.2 Frequency of testing. The frequency of conducting strength tests of concrete shall be as specified in Sections 1905.6.2.1 through 1905.6.2.4.

1905.6.2.1 Minimum frequency. For the first 250 cubic yards of each class of concrete placed each day, samples for strength tests of each such class shall be taken: (i) not less than once a day; (ii) not less than once for each 50 cubic yards (38 m³) of concrete; and (iii) not less than once for each 5,000 square feet (465 m²) of surface area for slabs or walls. After the first 250 cubic yards (191 m³), samples for strength tests of each class of concrete placed each day shall be taken once for each additional 150 cubic yards (115 m³). For structural concrete exposed to exposure class C2, additional field testing of the concrete for water content per AASHTO T 318 shall be required. At the discretion of the registered design professional of record, the frequency of testing may be reduced, but not less than once for each 150 cubic yards (115 m³).

1905.6.2.2 Minimum number. On a given project, if the total volume of concrete is such that the frequency of testing required by Section 1905.6.2.1 would provide less than five strength tests for a given class of concrete, tests shall be made from at least five randomly selected batches or from each batch if fewer than five batches are used.

1905.6.2.3 Small volume. When the total volume of concrete is less than 50 cubic yards (38 m³) on a given project, testing may be waived by the registered design professional of record.

1905.6.2.4 Strength test. A strength test shall be the average of the strengths of two 6 inch by 12 inch (152 mm by 3048 mm) or two 4 inch by 8 inch (102 mm by 203 mm) cylinders made from the same sample of concrete and tested at 28 days or at the test age designated for the determination of f'_c . Consideration shall be given specifying an f'_c at 56 days or later for concrete utilizing pozzolans in the mix proportions. At the discretion of the registered design professional of record or the approved agency performing the testing, additional pairs of test cylinders may be taken at the time of sampling for testing at a later date. The additional cylinders may be tested at a later date should the strength at the specified date not meet the required f'_c .

1905.6.3 Standard-cured specimens. Standard-cured specimens shall comply with the provisions of Sections 1905.6.3.1 through 1905.6.3.9.

†††**1905.6.3.1 Mix design.** The special inspector shall verify that the proportions indicated on the batch ticket for the concrete delivered to the construction site are as per the approved concrete mix design prior to concrete placement (see Table 1705.3 of this code). Concrete that does not meet the requirements of the approved concrete mix design shall be rejected.

1905.6.3.2 Sampling. Samples for strength tests shall be taken in accordance with ASTM C 172.

1905.6.3.3 Cylinders. Cylinders for strength tests shall be molded and standard-cured in accordance with ASTM C 31.

1905.6.3.3.1 On-site curing. The contractor shall be responsible for providing the specified field storage curing facility and for monitoring the temperature of the cylinders as defined in ASTM C 31.

1905.6.3.3.2 Final curing. Cylinders shall be placed in final curing conditions meeting the requirements of ASTM C 511. Cylinders shall be transported to final curing conditions by the licensed concrete testing laboratory within 48 hours of casting the cylinders.

1905.6.3.3.3 Compressive strength testing. The cylinders shall be tested in accordance with ASTM C 39.

1905.6.3.4 Evaluation of results. The strength level of an individual class of concrete shall be considered satisfactory if both of the following requirements are met:

1. Every arithmetic average of any three consecutive strength tests equals or exceeds f'_c .
2. No individual strength test (average of two cylinders) falls below f'_c by more than 500 psi (3.45 MPa) when f'_c is 5,000 psi (34.50 MPa) or less, or by more than 0.10 f'_c when f'_c is more than 5,000 psi (34.50 MPa).

1905.6.3.5 Correction. If either of the requirements of Section 1905.6.3.4 is not met, steps shall be taken to increase the average of subsequent strength test results. The requirements of Section 1905.6.5 shall govern if the requirement of Section 1905.6.3.4, Item 2, is not met.

1905.6.3.6 Field testing. Each time concrete is sampled for strength testing, the tests set forth in Items 1 through 5 shall be performed by the approved agency:

1. Temperature in accordance with ASTM C 1064.
2. Slump in accordance with ASTM C 143.
3. Unit weight in accordance with ASTM C 138.
4. Air content in accordance with ASTM C 173 or C 231. Testing in accordance with ASTM C 173 must be used for concretes made with lightweight aggregates, and other instances where ASTM C 231 is not applicable.
5. Water content per AASHTO T 318 when required by Section 1905.6.2.1 of this code.

1905.6.3.7 Mass concrete. For mass concrete sections identified as requiring monitoring, install temperature devices that are capable of measuring the temperature of the concrete continuously and record temperature data in increments that do not exceed 30 minutes for a minimum period of seven days per ACI 301, Section 8.3 or as otherwise directed by the registered design professional of record. In lieu of monitoring a concrete placement for core and outside face temperatures, a computer simulated thermal study may be substituted, if approved by the registered design professional of record. The computer simulated thermal study shall be of the largest section placed for a specific concrete mix proportion and reflect the thermal properties for the specific cementitious materials and quantities in the mix. The thermal study shall include the maximum temperature at which concrete may be placed and the ambient temperature range for concrete placement.

1905.6.3.8 Results of field tests. The approved agency shall immediately report the results of field tests performed to the contractor and concrete truck driver, or other representative of the concrete producer. If at any time during the concrete operations, the concrete is not in conformance with ASTM C 94, this code, or is otherwise compromised, it shall be rejected. If it is placed, the location of the load shall be recorded and a pair of cylinders shall be molded from the load per ASTM C 31 and included in the average f'_c . The approved agency shall keep a written record of all field tests at the construction site and make the records available upon request. These results need not be used in calculating the arithmetic average of consecutive test results required by Section 1905.6.3.4.

1905.6.3.9 Reporting requirements. Field sampling and testing shall be deemed a special inspection, and the approved agency performing the field sampling and testing of concrete at the construction site shall be subject to the requirements of Section 1704.1.1.2 regarding reporting of deficiencies.

1905.6.4 Field-cured specimens. Field-cured specimens shall comply with the provisions of Sections 1905.6.4.1 through 1905.6.4.4.

1905.6.4.1 When required. Where required by the commissioner, the results of strength tests of cylinders cured under field conditions shall be provided to the department.

1905.6.4.2 Curing. Field-cured cylinders shall be cured under field conditions in accordance with ASTM C 31.

1905.6.4.3 Sampling. Field-cured test cylinders shall be molded at the same time and from the same samples as laboratory-cured test cylinders.

1905.6.4.4 Correction. Procedures for protecting and curing concrete shall be improved when the strength of field-cured cylinders at the test age designated for determination of f'_c is less than 85 percent of that of companion laboratory-cured cylinders. The 85-percent limitation shall not apply if the field-cured strength exceeds f'_c by more than 500 psi (3.45 MPa).

1905.6.5 Low-strength test results. The investigation of low-strength test results shall be in accordance with the provisions of Sections 1905.6.5.1 through 1905.6.5.5.

1905.6.5.1 Precaution. If any strength test of laboratory-cured cylinders performed in accordance with Section 1905.6.2.4 falls below the specified value of f'_c by more than the values given in Section 1905.6.3.4, Item 2, or if tests of field-cured cylinders performed in accordance with Section 1905.6.4 indicate deficiencies in protection and curing, steps shall be taken to assure that the load-carrying capacity of the structure is not jeopardized.

1905.6.5.2 Core tests. Where calculations indicate that load-carrying capacity is significantly reduced, tests of cores drilled from the area in question in accordance with ASTM C 42 shall be permitted. In such cases, three cores shall be taken for each strength test that falls below the values given in Section 1905.6.3.4, Item 2.

1905.6.5.3 Condition of cores. Cores shall be prepared for transport and storage by wiping drilling water from their surfaces and placing the cores in water-tight bags or containers immediately after drilling. Cores shall be

CONCRETE

tested not earlier than 48 hours nor later than seven days after coring unless approved by the registered design professional of record.

1905.6.5.4 Test results. Concrete in an area represented by core tests shall be considered structurally adequate if the average of three cores is equal to at least 85 percent of f'_c and if no single core is less than 75 percent of f'_c . Additional testing of cores extracted from locations represented by erratic core strength results is permitted.

1905.6.5.5 Strength evaluation. If the criteria of Section 1905.6.5.4 are not met and the structural adequacy remains in doubt, the commissioner may order a strength evaluation in accordance with ACI 318, Chapter 27, for the questionable portion of the structure, or take other appropriate action. The registered design professional of record shall present to the commissioner a complete analysis showing the final safe load-carrying capacity of the questionable portion of the structure including any proposed remedial actions necessary for review and approval.

1905.6.6 Steel fiber-reinforced concrete. Steel fiber-reinforced concrete used in beams shall be subject to the requirements of ACI 318, Section 26.12.5. The required testing and inspection of steel fiber-reinforced concrete shall be deemed a special inspection. Steel fibers shall not be used in place of required reinforcing bars.

1905.7 Preparation of equipment and place of deposit. Preparation before concrete placement shall include the following:

1. Equipment for mixing and transporting concrete shall be clean.
2. Debris and ice shall be removed from spaces to be occupied by concrete.
3. Forms shall be properly coated.
4. Masonry filler units that will be in contact with concrete shall be well drenched.
5. Reinforcement shall be thoroughly clean of ice or other deleterious coatings.
6. Water shall be removed from the place of deposit before concrete is placed unless a tremie is to be used or unless otherwise permitted by the commissioner.
7. Laitance and other unsound material shall be removed before additional concrete is placed against hardened concrete.

1905.8 Mixing. Mixing of concrete shall be performed in accordance with Sections 1905.8.1 through 1905.8.3.

1905.8.1 General. Concrete shall be mixed until there is a uniform distribution of materials and shall be discharged completely before the mixer is recharged. Concrete delivered shall be proportioned in accordance with Section 1905.2.3. Modification to the approved concrete proportions shall be based upon standards acceptable to the commissioner.

1905.8.2 Ready-mixed concrete. Ready-mixed concrete shall be mixed and delivered in accordance with the requirements of ASTM C 94 or ASTM C 685.

1905.8.2.1 Concrete plants. Concrete plants shall be certified by the National Ready Mixed Concrete Association (NRMCA), the Concrete Industry Board Concrete Producer Certification Program, or other program acceptable to the department, and shall comply with the rules of the department. Concrete producers shall have their plants inspected quarterly and have their scales and trucks certified. In fulfilling this certification requirement, the concrete producer may present certification by either a New York City government agency, or by the New York State Department of Transportation subject to the approval of the commissioner.

1905.8.2.2 Delivery personnel. Concrete ready-mix truck drivers shall be certified by the NRMCA and shall comply with the rules of the department.

1905.8.2.3 Batch tickets. Batch tickets shall accompany every load of concrete delivered to a site. The batch ticket shall contain the information specified in Items 1 through 7:

1. Plant name and location.
2. Contract number and project.
3. Mix designation as to type and strength.
4. Each material in the load along with quantities of each by weight, and admixture by ounces or gallons.

5. The total amount of mix proportion water approved, quantity of water added at the plant and in transit, and the remaining water that can be added on site.
6. Any deviations from the approved concrete mix design and variations permitted by Section 1905.3.5.
7. The time the water was added to the batch.

1905.8.2.3.1 Copies of batch tickets. A copy of each batch ticket shall be given to the licensed concrete testing laboratory special inspector performing the acceptance testing at the delivery of each load. The licensed laboratory shall maintain copies of the batch tickets along with other inspection reports required by Section 1704.1, and make them available to the department upon request.

1905.8.3 Job-mixed concrete. Job-mixed concrete shall comply with ACI 318, Section 26.5.

1905.9 Conveying. The method and equipment for conveying concrete to the place of deposit shall comply with Sections 1905.9.1 and 1905.9.2.

1905.9.1 Method of conveyance. Concrete shall be conveyed from the mixer to the place of final deposit by methods that will prevent separation or loss of materials that may alter the properties of the concrete delivered. Cylinders shall be made to determine the quality of concrete. Such cylinders shall be made at the truck or at the point of placement of the concrete as determined by the special inspection agency or the registered design professional of record.

1905.9.2 Conveying equipment. The conveying equipment shall be capable of providing a supply of concrete at the site of placement without separation of ingredients and without interruptions sufficient to permit the loss of plasticity between successive increments.

1905.10 Depositing. The depositing of concrete shall comply with the provisions of Sections 1905.10.1 through 1905.10.8.

1905.10.1 Segregation. Concrete shall be deposited as nearly as practicable to its final position to avoid segregation due to rehandling or flowing.

1905.10.2 Placement timing. Concreting operations shall be carried on at such a rate that the concrete is at all times plastic and flows readily into spaces between reinforcement.

1905.10.3 Unacceptable concrete. Concrete that has partially hardened or been contaminated by foreign materials shall not be deposited in the structure.

1905.10.4 Retempering. Retempered concrete or concrete that has been remixed after initial set shall not be used.

1905.10.5 Continuous operation. After concreting has started, it shall be carried on as a continuous operation until placing of a panel or section, as defined by its boundaries or predetermined joints, is completed, except as permitted or prohibited by Section 1906.4.

1905.10.6 Placement in vertical lifts. The top surfaces of vertically formed lifts shall be generally level.

1905.10.7 Construction joints. When construction joints are required, they shall be made in accordance with Section 1906.4.

1905.10.8 Consolidation. Concrete shall be thoroughly consolidated by suitable means during placement and shall be thoroughly worked around reinforcement and embedded fixtures and into corners of the forms.

1905.11 Curing. The curing of concrete shall be in accordance with Sections 1905.11.1 through 1905.11.3.

1905.11.1 Regular. Concrete (other than high early strength) shall be maintained above 50°F (10°C) and in a moist condition for at least the first seven days after placement, except when cured in accordance with Section 1905.11.3.

1905.11.2 High-early-strength. High-early-strength concrete shall be maintained above 50°F (10°C) and in a moist condition for at least the first three days, except when cured in accordance with Section 1905.11.3.

1905.11.3 Accelerated curing. Accelerated curing of concrete shall comply with ACI 318, Section 26.5.3.2(c), but concrete temperature shall not exceed 160°F (71°C).

CONCRETE

1905.12 Cold weather requirements. Concrete that is to be placed during freezing or near-freezing weather shall comply with the following:

1. Adequate equipment shall be provided for insulating or heating concrete materials and protecting concrete during freezing or near-freezing weather.
2. Concrete materials and reinforcement, forms, fillers and ground with which concrete is to be in contact shall be free from frost. The temperature of surfaces in contact with fresh concrete, including but not limited to forms and reinforcing steel, shall be raised above freezing.
3. Frozen materials or materials containing ice shall not be used.

1905.13 Hot weather requirements. During hot weather, proper attention shall be given to ingredients, production methods, handling, placing, protection and curing to prevent excessive concrete temperatures or water evaporation that could impair the required strength or serviceability of the member or structure.

SECTION BC 1906 FORMWORK, EMBEDDED PIPES AND CONSTRUCTION JOINTS

1906.1 Formwork. The design, fabrication and erection of forms shall comply with the requirements of ACI 318, Section 26.11 and Section 3305.3 of this code.

1906.2 Removal of forms, shores and reshores. The removal of forms and shores, including slabs and beams (except where cast on the ground), and the installation of reshores shall comply with ACI 318, Section 26.11.2 and Section 3305.3 of this code.

1906.3 Conduits and pipes embedded in concrete. Conduits, pipes and sleeves of any material not harmful to concrete and within the limitations of ACI 318, Section 20.7, are permitted to be embedded in concrete with approval of the registered design professional of record.

1906.4 Construction joints. Construction joints, including their location, shall comply with the provisions of ACI 318, Section 26.5.6. Construction joints, including their location, shall be approved by the registered design professional of record prior to installation.

SECTION BC 1907 DETAILS OF REINFORCEMENT

1907.1 Hooks. Standard hooks on reinforcing bars used in concrete construction shall comply with ACI 318, Section 25.3.

1907.2 Minimum bend diameters. Minimum reinforcement bend diameters utilized in concrete construction shall comply with ACI 318, Section 25.3.2.

1907.3 Bending. The bending of reinforcement shall comply with Sections 1907.3.1 and 1907.3.2.

1907.3.1 Cold bending. Reinforcement shall be bent cold, unless otherwise permitted by the registered design professional of record.

1907.3.2 Embedded reinforcement. Reinforcement partially embedded in concrete shall not be field bent, except as shown on the construction documents or permitted by the registered design professional of record.

1907.4 Surface conditions of reinforcement. The surface conditions of reinforcement shall comply with the provisions of Sections 1907.4.1 through 1907.4.3.

1907.4.1 Coatings. At the time concrete is placed, reinforcement shall be free from mud, oil or other nonmetallic coatings that decrease bond. Zinc and epoxy-coating of steel reinforcement in accordance with ACI 318, Section 20.6.2 shall be permitted.

1907.4.2 Rust or mill scale. Except for prestressing reinforcement, steel reinforcement with rust, mill scale or a combination of both, shall be permitted, provided the minimum dimensions, including height of deformations and

weight of a hand-wire-brushed test specimen, comply with applicable ASTM specifications specified in Section 1903.6.

1907.4.3 Prestressing reinforcement. Prestressing reinforcement shall be clean and free of oil, dirt, scale, pitting and excessive rust. A light coating of rust is permitted.

1907.5 Placing reinforcement. The placement of concrete reinforcement shall comply with the provisions of Sections 1907.5.1 through 1907.5.4.

1907.5.1 Support. Reinforcement, including tendons, and posttensioning ducts shall be accurately placed and adequately supported before concrete is placed, and shall be secured against displacement within tolerances permitted in Section 1907.5.2. Where approved by the registered design professional of record, embedded items (such as dowels or inserts) that either protrude from precast concrete members or remain exposed for inspection are permitted to be embedded while the concrete is in a plastic state, provided the following conditions are met:

1. Embedded items are maintained in the correct position while the concrete remains plastic.
2. The concrete is properly consolidated around the embedded item. Embedded items are not required to be hooked or tied to reinforcement within the concrete.

1907.5.2 Tolerances. Unless otherwise specified by the registered design professional of record, reinforcement, including tendons, and posttensioning ducts shall be placed within the tolerances specified in Sections 1907.5.2.1 and 1907.5.2.2.

1907.5.2.1 d and cover. Tolerance for d and concrete cover shall be as required in ACI 318, Section 26.6.2.1(a).

1907.5.2.2 Bends and ends. Tolerance for longitudinal location of bends and ends of reinforcement shall be as required by ACI 318, Section 26.6.2.1(b).

1907.5.3 Welded wire reinforcement. Welded wire reinforcement with wire size not greater than W5 or D5 used in slabs not exceeding 10 feet (3048 mm) in span is permitted to be curved from a point near the top of the slab over the support to a point near the bottom of the slab at midspan, provided such reinforcement is either continuous over, or securely anchored at support.

1907.5.4 Welding. Welding of crossing bars shall not be permitted for assembly of reinforcement unless authorized by the registered design professional of record.

1907.6 Minimum spacing of reinforcement. The clear distance between reinforcing bars, bundled bars, tendons and ducts shall comply with ACI 318, Section 25.2.

1907.7 Concrete protection for reinforcement. The minimum concrete cover for reinforcement shall comply with Sections 1907.7.1 through 1907.7.8.

1907.7.1 Cast-in-place concrete (nonprestressed). Minimum specified concrete cover shall be provided for reinforcement in nonprestressed cast-in-place concrete construction in accordance with ACI 318, Section 20.6.1.3.1, but shall not be less than required by Sections 1907.7.6 and 1907.7.8 of this code.

1907.7.2 Cast-in-place concrete (prestressed). The minimum specified concrete cover for prestressed and nonprestressed reinforcement, ducts and end fittings in cast-in-place prestressed concrete shall comply with ACI 318, Section 20.6.1.3.2.

1907.7.3 Precast concrete (manufactured under plant control conditions). The minimum specified concrete cover for prestressed and nonprestressed reinforcement, ducts and end fittings in precast concrete manufactured under plant control conditions shall comply with ACI 318, Section 20.6.1.3.3.

1907.7.4 Bundled bars. The minimum specified concrete cover for bundled bars shall comply with ACI 318, Section 20.6.1.3.4.

1907.7.5 Headed shear stud reinforcement. For headed shear stud reinforcement, the minimum specified concrete cover shall comply with ACI 318, Section 20.6.1.3.5.

CONCRETE

1907.7.6 Corrosive environments. In corrosive environments or other severe exposure conditions, prestressed and nonprestressed reinforcement shall be provided with additional protection in accordance with ACI 318, Section 20.6.1.4.

1907.7.7 Future extensions. Exposed reinforcement, inserts and plates intended for bonding with future extensions shall be protected from corrosion.

1907.7.8 Fire protection. When this code requires a thickness of cover for fire protection greater than the minimum concrete cover in Section 1907.7, such greater thickness shall be specified on the construction documents.

1907.8 Special reinforcement details for columns. Offset bent longitudinal bars in columns and load transfer in structural steel cores of composite compression members shall comply with the provisions of ACI 318, Section 10.7.4.

1907.9 Connections. Connections between concrete framing members shall comply with the provisions of ACI 318, Chapters 15 and 16.

1907.10 Transverse reinforcement for compression members. Transverse reinforcement for compression members shall comply with the provisions of ACI 318, Section 25.7.2.

1907.11 Transverse reinforcement for flexural members. Transverse reinforcement for compression reinforcement in flexural members shall comply with the provisions of ACI 318, Section 25.7.1.

1907.12 Shrinkage and temperature reinforcement. Reinforcement for shrinkage and temperature stresses in concrete members shall comply with the provisions of ACI 318, Section 24.4.

1907.13 Requirements for structural integrity. The detailing of reinforcement and connections between concrete members shall comply with the provisions of ACI 318, Section 4.10 and Section 1912 of this code.

SECTION BC 1908 MODIFICATIONS TO ACI 318

1908.1 General. The text of ACI 318 shall be modified as indicated in Sections 1908.1.1 through 1908.1.10.

1908.1.1 ACI 318, Section 2.3. Modify existing definitions and add the following definitions to ACI 318, Section 2.3.

DESIGN DISPLACEMENT. Total lateral displacement expected for the design-basis earthquake, as specified by Section 12.8.6 of ASCE 7.

DETAILED PLAIN CONCRETE STRUCTURAL WALL. A wall complying with the requirements of Chapter 14 including 14.6.2.

ORDINARY PRECAST STRUCTURAL WALL. A precast wall complying with the requirements of ACI 318 except for Chapters 14, 17 and 18.

ORDINARY REINFORCED CONCRETE STRUCTURAL WALL. A cast-in-place wall complying with the requirements of ACI 318 except for Chapters 14, 17 and 18.

ORDINARY STRUCTURAL PLAIN CONCRETE WALL. A wall complying with the requirements of Chapter 14, excluding 14.6.2.

SPECIAL STRUCTURAL WALL. A cast-in-place or precast wall complying with the requirements of 18.2.4 through 18.2.8, 18.10 and 18.11, as applicable, in addition to the requirements for ordinary reinforced concrete structural walls or ordinary precast structural walls, as applicable. Where ASCE 7 refers to a “special reinforced concrete structural shear wall,” it shall be deemed to mean a “special structural wall.”

1908.1.2 ACI 318, Section 8.7.2. Modify ACI 318, Section 8.7.2.3 and 8.7.2.4, and add new sections 8.7.2.5 and 8.7.2.6 to read as follows:

8.7.2.3 – For prestressed slabs with Banded-Distributed tendon placement, the maximum spacing s of tendons or groups of tendons in the distributed direction shall be the lesser of $8h$ and 5 ft.

8.7.2.4 – The placement of tendons banded in both directions over the supports in two-way slabs is allowed. For Banded-Banded tendon placement, all provisions of ACI-318 Chapter 8 shall apply, except section 8.7.2.3, referring to the maximum tendon spacing limitation.

8.7.2.5 – At locations where the average effective prestress is not distributed, the two-way slab shall be designed using non-prestressed reinforcement.

8.7.2.6 – Concentrated Loads and openings shall be considered in determining tendon spacing.

1908.1.3 ACI 318, Section 14.1.4. Delete ACI 318, Section 14.1.4 and replace with the following:

14.1.4 – Plain concrete in structures assigned to Seismic Design Category C or D.

14.1.4.1 – Structures assigned to Seismic Design Category C or D shall not have elements of structural plain concrete, except as follows:

(a) Structural plain concrete basement, foundation or other walls below the base, as defined in ASCE 7, are permitted in detached one- and two-family dwellings three stories or less in height constructed with stud-bearing walls. In dwellings assigned to Seismic Design Category D, the height of the wall shall not exceed 8 feet (2438 mm), the thickness shall be not less than 7½ inches (190 mm), and the wall shall retain no more than 4 feet (1219 mm) of unbalanced fill. Walls shall have reinforcement in accordance with 14.6.1.

(b) Isolated footings of plain concrete supporting pedestals or columns are permitted, provided the projection of the footing beyond the face of the supported member does not exceed the footing thickness.

Exception: In detached one- and two-family dwellings three stories or less in height, the projection of the footing beyond the face of the supported member is permitted to exceed the footing thickness.

(c) Plain concrete footings supporting walls are permitted, provided the footings have at least two continuous longitudinal reinforcing bars. Bars shall not be smaller than No. 4 and shall have a total area of not less than 0.002 times the gross cross-sectional area of the footing. For footings that exceed 8 inches (203 mm) in thickness, a minimum of one bar shall be provided at the top and bottom of the footing. Continuity of reinforcement shall be provided at corners and intersections.

Exceptions:

1. In Seismic Design Categories A, B and C, detached one- and two-family dwellings three stories or less in height constructed with stud-bearing walls are permitted to have plain concrete footings without longitudinal reinforcement.
2. For foundation systems consisting of a plain concrete footing and a plain concrete stemwall, a minimum of one bar shall be provided at the top of the stem wall and at the bottom of the footing.
3. Where a slab on ground is cast monolithically with the footing, one No. 5 bar is permitted to be located at either the top of the slab or bottom of the footing.

1908.1.4 ACI 318, Section 14.6. Modify ACI 318, Section 14.6 by adding new Section 14.6.2 to read as follows:

14.6.2 – Detailed plain concrete structural walls.

14.6.2.1 – Detailed plain concrete structural walls are walls conforming to the requirements of ordinary structural plain concrete walls and 14.6.2.2.

14.6.2.2 – Reinforcement shall be provided as follows:

- (a) Vertical reinforcement of at least 0.20 square inch (129 mm²) in cross-sectional area shall be provided continuously from support to support at each corner, at each side of each opening and at the ends of walls. The continuous vertical bar required beside an opening is permitted to substitute for one of the two No. 5 bars required by 14.6.1.
- (b) Horizontal reinforcement at least 0.20 square inch (129 mm²) in cross-sectional area shall be provided:
 1. Continuously at structurally connected roof and floor levels and at the top of walls;
 2. At the bottom of load-bearing walls or in the top of foundations where doweled to the wall; and

CONCRETE

3. At a maximum spacing of 120 inches (3048 mm).

Reinforcement at the top and bottom of openings, where used in determining the maximum spacing specified in Item 3 above, shall be continuous in the wall.

1908.1.5 ACI 318, Section 15.3. Modify ACI 318 by adding Section 15.3.2 to read as follows:

15.3.2 When the specified compressive strength of concrete in a column is greater than 1.4 times that specified for a floor system, the following additional requirements shall be adhered to:

1. All of the design provisions of Section 15.3 (unmodified) are adhered to.
2. The concrete construction is supervised and inspected continuously by a full-time professional engineer responsible for the concrete placement special inspection. Such professional engineer shall not delegate this responsibility to any subordinates.

1908.1.6 ACI 318, Section 17.2.3. Modify ACI 318 Sections 17.2.3.4.2, 17.2.3.4.3(d) and 17.2.3.5.2 to read as follows:

17.2.3.4.2 – Where the tensile component of the strength-level earthquake force applied to anchors exceeds 20 percent of the total factored anchor tensile force associated with the same load combination, anchors and their attachments shall be designed in accordance with 17.2.3.4.3. The anchor design tensile strength shall be determined in accordance with 17.2.3.4.4.

Exception: Anchors designed to resist wall out-of-plane forces with design strengths equal to or greater than the force determined in accordance with ASCE 7 Equation 12.11-1 or 12.14-10 shall be deemed to satisfy Section 17.2.3.4.3(d).

17.2.3.4.3(d) – The anchor or group of anchors shall be designed for the maximum tension obtained from design load combinations that include E, with E increased by Ω_0 . The anchor design tensile strength shall be calculated from 17.2.3.4.4.

17.2.3.5.2 – Where the shear component of the strength-level earthquake force applied to anchors exceeds 20 percent of the total factored anchor shear force associated with the same load combination, anchors and their attachments shall be designed in accordance with 17.2.3.5.3. The anchor design shear strength for resisting earthquake forces shall be determined in accordance with 17.5.

Exceptions:

1. For the calculation of the in-plane shear strength of anchor bolts attaching wood sill plates of bearing or nonbearing walls of light-frame wood structures to foundations or foundation stem walls, the in-plane shear strength in accordance with 17.5.2 and 17.5.3 need not be computed and 17.2.3.5.3 shall be deemed to be satisfied provided all of the following are met:
 - 1.1. The allowable in-plane shear strength of the anchor is determined in accordance with AWC NDS Table 12E for lateral design values parallel to grain.
 - 1.2. The maximum anchor nominal diameter is $5/8$ inch (16 mm).
 - 1.3. Anchor bolts are embedded into concrete a minimum of 7 inches (178 mm).
 - 1.4. Anchor bolts are located a minimum of $1\frac{3}{4}$ inches (45 mm) from the edge of the concrete parallel to the length of the wood sill plate.
 - 1.5. Anchor bolts are located a minimum of 15 anchor diameters from the edge of the concrete perpendicular to the length of the wood sill plate.
 - 1.6. The sill plate is 2-inch (51 mm) or 3-inch (76 mm) nominal thickness.
2. For the calculation of the in-plane shear strength of anchor bolts attaching cold-formed steel track of bearing or nonbearing walls of light-frame construction to foundations or foundation stem walls, the in-plane shear strength in accordance with 17.5.2 and 17.5.3 need not be computed and 17.2.3.5.3 shall be deemed to be satisfied provided all of the following are met:
 - 2.1. The maximum anchor nominal diameter is $5/8$ inch (16 mm).
 - 2.2. Anchors are embedded into concrete a minimum of 7 inches (178 mm).

- 2.3. Anchors are located a minimum of $1\frac{3}{4}$ inches (45 mm) from the edge of the concrete parallel to the length of the track.
- 2.4. Anchors are located a minimum of 15 anchor diameters from the edge of the concrete perpendicular to the length of the track.
- 2.5. The track is 33 to 68 mil (0.84 mm to 1.73 mm) designation thickness.

Allowable in-plane shear strength of exempt anchors, parallel to the edge of concrete, shall be permitted to be determined in accordance with AISI S100, Section J3.3.1.

- 3. In light-frame construction bearing or non-bearing walls, shear strength of concrete anchors less than or equal to 1 inch (25 mm) in diameter attaching sill plate or track to foundation or foundation stem wall need not satisfy 17.2.3.5.3(a) through (c) when the design strength of the anchors is determined in accordance with 17.5.2.1(c).

1908.1.7 ACI 318, Section 18.2.1. Modify ACI 318 Sections 18.2.1.2 and 18.2.1.6 to read as follows:

18.2.1.2 – Structures assigned to Seismic Design Category A shall satisfy requirements of Chapters 1 through 17 and 19 through 26; Chapter 18 does not apply. Structures assigned to Seismic Design Category B, C, or D also shall satisfy 18.2.1.3 through 18.2.1.7, as applicable. Except for structural elements of plain concrete complying with Section 1908.1.2 of the *New York City Building Code*, structural elements of plain concrete are prohibited in structures assigned to Seismic Design Category C or D.

18.2.1.6 - Structural systems designated as part of the seismic force-resisting system shall be restricted to those permitted by ASCE 7 as modified by Chapter 16 of the *New York City Building Code*. Except for Seismic Design Category A, for which Chapter U8 does not apply, the following provisions shall be satisfied for each structural system designated as part of the seismic force-resisting system, regardless of the Seismic Design Category:

- (a) Ordinary moment frames shall satisfy 18.3.
- (b) Ordinary reinforced concrete structural walls and ordinary precast structural walls need not satisfy any provisions in Chapter 18.
- (c) Intermediate moment frames shall satisfy 18.4.
- (d) Intermediate precast structural walls shall satisfy 18.5.
- (e) Special moment frames shall satisfy 18.6 through 18.9.
- (f) Special structural walls shall satisfy 18.10.
- (g) Special structural walls constructed using precast concrete shall satisfy 18.11.

All special moment frames and special structural walls shall also satisfy 18.2.4 through 18.2.8.

1908.1.8 ACI 318, Section 18.5. Modify ACI 318, Section 18.5, by adding new Section 18.5.2.2 and renumbering existing Sections 18.5.2.2 and 18.5.2.3 to become 18.5.2.3 and 18.5.2.4, respectively.

18.5.2.2 – Connections that are designed to yield shall be capable of maintaining 80 percent of their design strength at the deformation induced by the design displacement or shall use Type 2 mechanical splices.

18.5.2.3 – Elements of the connection that are not designed to yield shall develop at least $1.5 S_y$.

18.5.2.4 – In structures assigned to Seismic Design Category D, wall piers shall be designed in accordance with Section 18.10.8 or 18.14 in ACI 318.

1908.1.9 ACI 318, Section 18.11. Modify ACI 318, Section 18.11.2 to read as follows:

18.11.2.1 – Special structural walls constructed using precast concrete shall satisfy all the requirements of Section 18.10 for cast-in-place special structural walls in addition to Section 18.5.2.

1908.1.10 ACI 318, Section 18.13.1.1. Modify ACI 318, Section 18.13.1.1, to read as follows:

18.13.1.1 Foundations resisting earthquake-induced forces or transferring earthquake-induced forces between a structure and ground shall comply with the requirements of Section 18.13 and other applicable provisions of ACI 318 unless modified by Chapter 18 of the *New York City Building Code*.

1908.1.11 ACI 318, Section 24.2.3.5. Add the following to Section 24.2.3.5:

CONCRETE

When calculating M_a , consideration shall be given to loads imposed during construction.

SECTION BC 1909 STRUCTURAL PLAIN CONCRETE

1909.1 Scope. The design and construction of structural plain concrete, both cast-in-place and precast, shall comply with the minimum requirements of ACI 318, as modified by Section 1908 of this chapter.

Exception: For Group R-3 occupancies and buildings of other occupancies less than two stories above grade plane of light-frame construction, the required footing thickness of ACI 318 is permitted to be reduced to 6 inches (152 mm), provided that the footing does not extend more than 4 inches (102 mm) on either side of the supported wall.

SECTION BC 1910 MINIMUM SLAB PROVISIONS

1910.1 General. The thickness of concrete floor slabs supported directly on the ground shall not be less than 3½ inches (89 mm). A 6-mil (0.006 inch; 0.15 mm) polyethylene vapor retarder with joints lapped not less than 6 inches (152 mm) shall be placed between the base course or subgrade and the concrete floor slab, or other approved equivalent methods or materials shall be used to retard vapor transmission through the floor slab.

Exception: A vapor retarder is not required:

1. For detached structures accessory to occupancies in Group R-3, such as garages, utility buildings or other unheated facilities.
2. For unheated storage rooms having an area of less than 70 square feet (6.5 m²) and carports attached to occupancies in Group R-3.
3. For buildings of other occupancies where migration of moisture through the slab from below will not be detrimental to the intended occupancy of the building.
4. For drive ways, walks, patios and other flat work that will not be enclosed at a later date.
5. Where approved based on local site conditions.

SECTION BC 1911 SHOTCRETE

1911.1 General. Shotcrete is mortar or concrete that is pneumatically projected at high velocity onto a surface. Except as specified in this section, shotcrete shall conform to the requirements of this chapter for plain or reinforced concrete.

1911.1.1 Qualifications. Nozzlemen shall be ACI certified.

1911.2 Proportions and materials. Shotcrete proportions shall be selected that allow suitable placement procedures using the delivery equipment selected and shall result in finished in-place hardened shotcrete meeting the strength requirements of this code. Prior to using shotcrete, the mix design for shotcrete shall be performed by the contractor and shall be approved by the registered design professional of record. The approved mix design shall be filed with the department based upon preconstruction tests performed in accordance with Section 1911.5.

1911.3 Aggregate. Coarse aggregate, if used, shall not exceed ¾ inch (19.1 mm).

1911.4 Reinforcement. Reinforcement used in shotcrete construction shall comply with the provisions of Sections 1911.4.1 through 1911.4.4.

1911.4.1 Size. The maximum size of reinforcement shall be No. 5 bars unless it is demonstrated by preconstruction tests that adequate encasement of larger bars will be achieved.

1911.4.2 Clearance. When No. 5 or smaller bars are used, there shall be a minimum clearance between parallel reinforcement bars of 2½ inches (64 mm). When bars larger than No. 5 are permitted, there shall be a minimum

clearance between parallel bars equal to six diameters of the bars used. When two curtains of steel are provided, the curtain nearer the nozzle shall have a minimum spacing equal to 12 bar diameters and the remaining curtain shall have a minimum spacing of six bar diameters.

Exception: Subject to the approval of the commissioner, required clearances shall be reduced where it is demonstrated by preconstruction tests that adequate encasement of the bars used in the design will be achieved.

1911.4.3 Splices. Lap splices of reinforcing bars shall utilize the noncontact lap splice method with a minimum clearance of 2 inches (51 mm) between bars. The use of contact lap splices necessary for support of the reinforcing is permitted when approved by the commissioner, based on satisfactory preconstruction tests that show that adequate encasement of the bars will be achieved, and provided that the splice is oriented so that a plane through the center of the spliced bars is perpendicular to the surface of the shotcrete.

1911.4.4 Spirally tied columns. Shotcrete shall not be applied to spirally tied columns.

1911.5 Preconstruction tests. A test panel shall be shot by each nozzleman and then, cured, cored or sawn, examined and tested prior to commencement of the project. The sample panel shall be representative of the project and simulate job conditions as closely as possible. The panel thickness and reinforcing shall reproduce the thickest and most congested area specified in the structural design. It shall be shot for each proposed concrete mix, at the proposed angles. The equipment used in preconstruction testing shall be the same equipment used in the work requiring such testing, unless substitute equipment is approved by the commissioner. Cores taken from the test panel shall be used to demonstrate that the proposed mix design meets the requirements of the construction documents and this code. All sampling and testing of shotcrete shall be performed by a licensed concrete testing laboratory. Construction of the test panels shall be witnessed by the special inspection agency responsible for the shotcrete construction.

1911.5.1 Sampling and testing. Six core samples shall be obtained from each preconstruction test panel:

1. Three specimens with no reinforcing shall be sampled and prepared in accordance with ASTM C 42; and
2. Three specimens with reinforcing steel.

1911.5.2 Compressive strength testing. The licensed concrete testing laboratory that sampled the preconstruction test panels shall test the core specimens with no reinforcing in accordance with ASTM C 39.

1911.5.3 Evaluation of strength tests. The shotcrete test panel strength shall be deemed acceptable if the mean compressive strength of the set of three cores shall equal or exceed f'_c with no individual core less than $0.75 f'_c$.

1911.5.4 Visual grading of cores. Core samples and/or test panels shall be cut with reinforcing steel and shall be visually evaluated by the registered design professional of record for the structural design, or a registered design professional acceptable to the registered design professional of record for the structural design. The nozzleman shall not proceed with work if cores and/or sections cut show the rebar not to be encapsulated with concrete to the satisfaction of the registered design professional of record for the structural design.

1911.6 Rebound. Any rebound or accumulated loose aggregate shall be removed from the surfaces to be covered prior to placing the initial or any succeeding layers of shotcrete. Rebound shall not be used as aggregate.

1911.7 Joints. Except where permitted herein, unfinished work shall not be allowed to stand for more than 30 minutes unless edges are sloped to a thin edge. For structural elements that will be under compression and for construction joints shown on the approved construction documents, square joints are permitted. Before placing additional material adjacent to previously applied work, sloping and square edges shall be cleaned and wetted.

1911.8 Damage. In-place shotcrete that exhibits sags, sloughs, segregation, honeycombing, sand pockets or other obvious defects shall be removed and replaced. Shotcrete above sags and sloughs shall be removed and replaced while still plastic.

1911.9 Curing. During the curing periods specified herein, shotcrete shall be maintained above 40°F (4°C) and in moist condition.

1911.9.1 Initial curing. Shotcrete shall be kept continuously moist for 24 hours after shotcreting is complete or shall be sealed with an approved curing compound.

CONCRETE

1911.9.2 Final curing. Final curing shall continue for seven days after shotcreting, or for three days if high-early-strength cement is used, or until the specified strength is obtained. Final curing shall consist of the initial curing process or the shotcrete shall be covered with an approved moisture-retaining cover.

1911.9.3 Natural curing. Natural curing shall not be used in lieu of that specified in this section unless the relative humidity remains at or above 85 percent, and is authorized by the registered design professional of record and approved by the commissioner.

1911.10 Strength tests. Strength tests for shotcrete shall be made by an approved agency on specimens that are representative of the work, and that are sampled and prepared in accordance with ASTM C 42.

1911.10.1 Sampling. Specimens shall be taken from the in-place work or from test panels, and shall be taken at least once each shift, but not less than three cores for each 50 cubic yards (38.2 m³) of shotcrete. With the approval of the registered design professional of record, when in-place cores are not feasible due to rebar spacing, test panels shall be shot in the same position as the work represented. One panel shall be cast for each 50 cubic yards (38.2 m³) of concrete and 3 cores shall be taken per ASTM C 42.

1911.10.2 Panel criteria. When the maximum-size aggregate is larger than $\frac{3}{8}$ inch (9.5 mm), the test panels shall have minimum dimensions of 18 inches by 18 inches (457 mm by 457 mm). When the maximum-size aggregate is $\frac{3}{8}$ inch (9.5 mm) or smaller, the test panels shall have minimum dimensions of 12 inches by 12 inches (305 mm by 305 mm). Panels shall be shot in the same position as the work, during the course of the work and by the nozzlemen doing the work. The conditions under which the panels are cured shall be the same as the work.

1911.10.3 Acceptance criteria. The average compressive strength of three cores from the in-place work tested in accordance with ASTM C 39 shall equal or exceed $0.85 f'_c$ with no single core less than $0.75 f'_c$. The average compressive strength of three cubes taken from the in-place work or a single test panel shall equal or exceed f'_c with no individual cube less than $0.88 f'_c$. To check accuracy, locations represented by erratic core or cube strengths shall be retested.

SECTION BC 1912 STRUCTURAL INTEGRITY REQUIREMENTS

1912.1 General. Reinforced concrete structures shall meet all the requirements of Sections 1912.1 through 1912.3. Concrete slabs on metal deck shall be governed by the provisions of Chapter 22. Reinforcement provided for gravity, seismic and wind forces or for other purposes may be regarded as forming part of, or the whole of, these requirements. Reinforcing provided for one requirement may be counted towards the other requirements.

1912.2 Continuity and ties. The structural integrity requirements of ACI 318, Sections 4.10 and 8.7.4.2, shall apply. In addition, the following requirements shall be met.

1912.2.1 Slab reinforcement. At all floor and roof levels, slabs shall have a mat of bottom reinforcement in two perpendicular (or roughly perpendicular) directions. Reinforcement in this bottom mat shall be made continuous with lap, mechanical or welded tension splices.

1912.2.1.1 Bottom mat reinforcement. In each direction, the bottom mat reinforcement shall be not less than, the steel required for temperature reinforcement. The bottom mat reinforcement shall be anchored at discontinuous edges within the column strip, reentrant corners, elevation changes and anywhere else the continuity of the reinforcing is interrupted.

Exception: Flat plate middle strip bottom mat reinforcing perpendicular to discontinuous slab edges. In addition, the main bottom mat reinforcement in one-way slabs shall be anchored at discontinuous edges.

1912.2.2 Peripheral ties. At each floor and roof level, reinforcement forming a continuous peripheral tie shall be provided. Peripheral ties shall be located within perimeter beams or walls, where they occur, or within 4 feet (1219 mm) of the edge of slab, where perimeter beams or walls do not occur. Continuous tie reinforcement shall be equal to half of the bottom reinforcement within the edge or edge strip for two-way slabs but not less than two bars.

1912.2.3 Horizontal ties. At each column, beam reinforcement or slab bottom reinforcement shall be provided at each level that can develop a tension force equal to the maximum of Item 1 or 2:

1. Three times the load entering the column at that level, using a load combination of $1.0 \times DL$ (self weight of structure only).
2. One and a half times the load entering the column at that level using the load combinations of $(1.2 DL + 1.6 LL)$ or $1.4 DL$.
3. For transfer elements only, in lieu of Item 1 or 2, the horizontal reinforcement shall be anchored at all supports.

1912.2.3.1 Bottom reinforcing. This beam or slab bottom reinforcement shall be distributed around the column perimeter and shall be extended on all sides of the column into the adjacent slab for at least one-third of the span length. Where reinforcing bars cannot be extended beyond the column (e.g., at slab edges and openings), they shall be hooked or otherwise developed within the column.

1912.2.4 Vertical ties. Each column and each wall carrying vertical load shall be vertically tied continuously from its lowest to highest level. The vertical ties composed of vertical column reinforcement shall be capable of resisting a tensile force equal to the maximum design dead and live load received by the column or wall from any one story within four floors below. A concrete compression element subjected to load reversal may have a reinforcing bar splice length equal to the tension splice length required by ACI 318 reduced by the ratio of $A_s \text{ required} / A_s$, provided but not less than the splice required for compression.

1912.3 Precast concrete general. Precast concrete structural elements shall be reinforced to meet all of the requirements of this section. However, reinforcement provided for gravity, seismic and wind forces and for other purposes may be regarded as forming part of, or the whole of, these requirements. Reinforcing provided for one requirement may be counted towards the other requirements.

1912.3.1 Continuity and ties. The structural integrity requirements of ACI 318, Section 4.10, shall apply. In precast and composite structures, ties within precast structural elements shall be continuous and shall be anchored to the supporting structure. In addition to Sections 1912.2.2 and 1912.2.4, the following requirements shall be met.

1912.3.1.1 End connections. End connections of all precast slabs, beams and girders shall have an axial tension capacity equal to the larger of the vertical shear capacity of the connection at either end, or at least 2 percent of the maximum factored vertical dead and live load in the precast compression element, whichever is larger, but not less than 20 kips or 2,500 pounds per linear foot of slab (36.48 kN/m). Where more than one element frames in one direction, none of the elements or connections shall have an axial tension capacity of less than 1 percent of the column load but not less than 20 kips.

1912.3.1.2 Side connections. Side connections of all precast elements shall have an axial tension capacity not less than the steel required for temperature reinforcement of the larger element at either side.

1912.3.1.3 Connection forces. For design of the connections, the transverse shear force and the axial tensile force need not be considered to act simultaneously.

1912.3.2 Joints. Joints in precast structures shall not rely on friction due to gravity to transfer load.

1912.3.3 Bearing. The net bearing area shall not be less than 2 inches (51 mm) wide and 3 inches (76 mm) long in the direction of the member.

SECTION BC 1913 ASPHALT PAVING

1913.1 Definitions. The following terms are defined in Chapter 2.

ASPHALT.

ASPHALTIC CONCRETE or ASPHALT PAVING.

I-4 MIX.

RECLAIMED ASPHALT PAVEMENT.

CONCRETE

1913.2 Reclaimed asphalt pavement content in asphaltic concrete. Asphaltic concrete, other than I-4 mix or other approved heavy duty asphaltic concrete mix, shall contain not less than 30 percent reclaimed asphalt pavement, as measured by weight. I-4 mix or other approved heavy duty asphaltic concrete mix shall contain not less than 10 percent reclaimed asphalt pavement, as measured by weight. Reclaimed asphalt paving used in asphaltic concrete shall comply with ASTM D692 or ASTM D1073.

Exceptions:

1. Asphaltic concrete used in a project where the content of asphaltic concrete is governed by a federal or state law, rule, regulation, guideline, or specification that requires a different composition.
2. Asphaltic concrete used for runways, taxiways, or other surfaces utilized by aircraft.
3. The commissioner may waive compliance with this section if the commissioner, after consulting with the commissioner of transportation and the owners or persons in charge of all asphalt plants located within the city, finds that a sufficient supply of reclaimed asphalt pavement is not available.